Kepler’s laws

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use orbital shape, equal areas and period comparisons.

SUCCESS CRITERIA - I can...

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A launch has v_x=5 m/s and v_y, i=30 m/s upward. Use g=10 m/s². Find time to the top and total time back to launch height.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

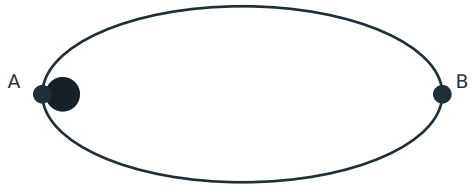
Worked reasoning: For an ideal Sun orbit, a=4 AU gives $T=\sqrt{(4^3)}=8$ years. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

Your turn: Draw an ellipse, put the Sun at a focus and mark the faster and slower parts.

Marzano check - circle: ① ② ③

03 Look and explain

Mark the Sun at a focus on an ellipse. Label where the object is fastest and slowest.



Marzano check - circle: ① ② ③

04 Build the solution

Supplied comparisons: a=1 AU givesT=1 year; a=4 AU givesT=8 years. Which orbit takes longer? How many Earth years for two orbits of the outer object?

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? What else is needed?

Marzano check - circle: 1 2 3

06 Choose a method

A ball is dropped from rest and falls ideally for 5 s. Use g=10 m/s². Find its speed and direction.

Marzano check - circle: 1 2 3

07 Exit check

An asteroid is nearest the Sun at point A and farthest at B. At which point is it fastest? Name which law helps explain this.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two Sun orbits have supplied periods 1 and 8 years. Their semimajor axes are 1 and 4 AU. Explain why four times the size does not mean four times the period.

Marzano check - circle: 1 2 3